

Results on Hexagonal Tile Rewriting Grammars

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Abstract. Tile rewriting grammars are a new model for defining picture languages. In this paper we propose hexagonal tile rewriting grammars (HTRG) for generating hexagonal picture languages. Closure properties of HTRG are proved for some basic operations. We compare HTRG with hexagonal tiling systems.

Keywords: Hexagonal tile rewriting grammars, picture languages, Hexagonal tiling systems and locally testable languages.

1 Introduction

Image generation in formal languages can be done in many ways. For example weighted finite automata, cellular automata are helpful to compress images and also to generate them. Using grammar system images can be generated by rewriting rules. Parallel and sequential array grammars were also used to generate images. In studies of hexagonal picture languages several grammars were introduced in the literature to generate various classes of pictures. One such grammar is the hexagonal kolam array grammar (HKAG) by Siromoneys [9] generating a class of hexagonal pictures (arrays) (HKAL). Based on the study [9], Subramanian [10] considered a hierarchy of hexagonal array grammars (HAG) generating classes of hexagonal arrays on triangular grids (HAL). Recently Dersanambika et al. [5] have defined local and recognizable hexagonal picture languages (HLOC and HREC). Tile rewriting grammars (TRG) are a new model for defining picture languages [2,3,4], combining the rewriting rules with the tiling system [7]. A TRG rule is a scheme having a nonterminal symbol to the left and a local language to the right over terminals and nonterminals; that is, the right part is specified by a set of fixed size tiles [3], the size of a TRG rule need not be fixed. To our knowledge, this approach is novel and is able to generate an interesting gamut of pictures: grids, spirals, and in particular a language of nested frames, which is in some way the analogue of a Dyck language [3]. For other interesting studies of rectangular, hexagonal and triangular arrays, we refer to [1,6,8,12].

In this paper, we introduce hexagonal tile rewriting grammars and generate interesting images. We compare the class HTRG with hexagonal tiling systems.

We prove that this model has greater generative capacity than the hexagonal tiling systems.

2 Preliminaries

Let Σ be a finite alphabet of symbols. A hexagonal picture p over Σ is a hexagonal array of symbols of Σ .

The set of all hexagonal arrays over the alphabet Σ is denoted by Σ^{**H} . A hexagonal picture language L over Σ is a subset of Σ^{**H} .

Let $\Sigma^{(\ell, m, n), H}$ be the set of hexagonal pictures of size (ℓ, m, n) .

For the notions of size of a hexagonal picture, hexagonal subpictures, hexagonal tile, local hexagonal picture languages ($HLOC$), recognizable hexagonal picture languages ($HREC$), hexagonal tiling systems (HTS), tiling recognizable hexagonal picture languages ($\mathcal{L}(HTS)$), we refer to [11] which has appeared in this volume.

3 Hexagonal Tile Rewriting Grammars and Hexagonal Picture Languages

In this section, we introduce some basic definitions, hexagonal tile rewriting grammars with an example and obtain a basic property.

Definition 1. Let p be a hexagonal picture of size (ℓ, m, n) . A subpicture of p at position (i, j, k) is a picture q such that, if (ℓ', m', n') is the size of q , then $\ell' \leq \ell$, $m' \leq m$, $n' \leq n$. We denote it by $q \trianglelefteq_{(i, j, k)} p$, or $q \trianglelefteq p \equiv \exists i, j, k (q \trianglelefteq_{(i, j, k)} p)$. Moreover, if $q \trianglelefteq_{(i, j, k)} p$, we define $coor_{(i, j, k)}(q, p)$ as the set of coordinates of p where q is located. Conventionally, $coor_{(i, j, k)}(q, p) = \phi$, if q is not a subpicture of p . If q coincides with p , we write $coor(p)$ instead of $coor_{(1, 1, 1)}(p, p)$.

Example 1. Let p be a hexagonal picture of size $(3, 3, 3)$ and let q be a subpicture of p at position $(1, 2, 1)$ of size $(2, 2, 2)$. Then the $coor(1, 2, 1)(q, p) = \{(1, 2, 1), (1, 2, 2), \dots, (2, 3, 2)\}$.

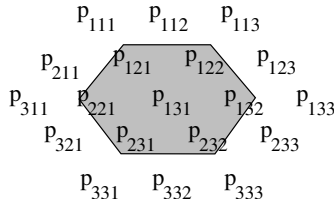


Fig. 1. Subpicture of a hexagonal picture

Definition 2. Let γ be an equivalence relation on $\text{coor}(p)$, written $(x, y, z) \stackrel{\gamma}{\sim} (x', y', z')$. Two subpictures $q \trianglelefteq_{(i,j,k)} p$, $q' \trianglelefteq_{(i',j',k')} p$ are γ -equivalent, written $q \stackrel{\gamma}{\sim} q'$, iff for all pairs $(x, y, z) \in \text{coor}_{(i,j,k)}(q, p)$ and $(x', y', z') \in \text{coor}_{(i',j',k')}(q', p)$ it holds $(x, y, z) \stackrel{\gamma}{\sim} (x', y', z')$.

A homogeneous C -subpicture $q \trianglelefteq p$ is called maximal with respect to relation γ iff for every γ -equivalent C -subpicture q' we have,

$$\text{coor}(q, p) \cap \text{coor}(q', p) = \emptyset \quad \text{or} \quad \text{coor}(q', p) \subseteq \text{coor}(q, p)$$

Definition 3. For a picture $p \in \Sigma^{**H}$, the set of subpictures (or tiles) with size (g, h, k) is

$$B_{g,h,k}(p) = \{q \in \Sigma^{(g,h,k)H} / q \trianglelefteq p\}.$$

Definition 4. Consider a set of tiles $\omega \subseteq \Sigma^{(i,j,k)H}$. The locally testable hexagonal language in the strict sense defined by ω (written $HLOC_u(\omega)$, ‘ u ’ stands for unbordered picture) is the set of pictures $p \in \Sigma^{**H}$ such that $B_{(i,j,k)}(p) \subseteq \omega$.

The locally testable hexagonal language defined by a finite set of tiles $HLOC_{u,eq}(\{\omega_1, \omega_2, \dots, \omega_n\})$ ‘ eq ’ stands for equality test) is the set of pictures $p \in \Sigma^{**H}$ such that for some x , $B_{i,j,k}(p) = \omega_x$.

The bordered locally testable hexagonal language defined by a finite set of tiles $HLOC_{eq}(\{\omega_1, \omega_2, \dots, \omega_n\})$ is the set of pictures $p \in \Sigma^{**H}$ such that for some x , $B_{i,j,k}(\hat{p}) = \omega_x$.

Definition 5 (Substitution). If p, q, q' are hexagonal pictures, $q \trianglelefteq_{(i,j,k)} p$, and q, q' have the same size, then $p[q'/q]_{(i,j,k)}$ denotes the hexagonal picture obtained by replacing the occurrence of q at position (i, j, k) in p with q' .

Hexagonal Tile Rewriting Grammars

Definition 6. A hexagonal tile rewriting grammar (HTRG) is a tuple $(\Sigma \cup \{b\}, N, S, R)$, where Σ is the terminal alphabet, $\{b\}$ is the blank symbol, N is a set of nonterminal symbols, $S \in N$ is the starting symbol, R is a set of rules. R may contain two kinds of rules,

Type 1 : $A \rightarrow t$, where $A \in N$, $t \in (\Sigma \cup \{b\} \cup N)^{(g,h,k)H}$ with $g, h, k > 0$

Type 2 : $A \rightarrow \omega$, where $A \in N$, $\omega \subseteq \{t/t \in (\Sigma \cup \{b\} \cup N)^{(x,y,z)H}\}$ with $x, y, z > 0$

Notice that type 1 is not a special case of type 2. Intuitively a rule of type 1 is intended to match a subpicture of small bounded size, identical to the right part t . A rule of type 2 matches any subpicture of any size which can be tiled using all the elements t of the set ω . The blank symbol ‘ b ’ is introduced in HTRG rules in order to generate arrow head hexagons of six types.

Definition 7. Consider a hexagonal tile rewriting grammar $G = (\Sigma \cup \{b\}, N, S, R)$. Let $p, p' \in (\Sigma \cup \{b\} \cup N)^{(\ell,m,n)H}$ be pictures of identical size, and let γ, γ' be equivalence relations over $\text{coor}(p)$. We say that (p', γ') is derived in one step from (p, γ) , written $(p, \gamma) \Rightarrow_G (p', \gamma')$ iff for some $A \in N$ and for some rule $\rho : A \rightarrow \dots \in R$ there exists in p a A -subpicture $r \trianglelefteq_{(i,j,k)} p$, maximal with respect to γ , such that

(i) p' is obtained substituting r with a picture s (i.e.,) $p' = p[s/r]_{(i,j,k)}$ where ' s ' is defined as follows:

Type 1 : Fixed size : if $\rho = A \rightarrow t$, then $s = t$

Type 2 : Variable size : if $\rho = A \rightarrow \omega$, then $s \in HLOC_{u,eq(\omega)}$.

(ii) Let z be $coord(i, j, k)$. Let Γ be the γ -equivalence class containing z . Then, γ' is equal to γ , for all the equivalence classes $\neq \Gamma$; Γ in γ' is divided in two equivalence classes, z and its complement with respect to Γ ($= \phi$ if $z = \Gamma$). More formally,

$$\gamma' = \gamma \setminus \{(x_1, y_1, z_1), (x_2, y_2, z_2) / (x_1, y_1, z_1) \in z \text{ or } (x_2, y_2, z_2) \in z\}.$$

The subpicture r is named the application area of rule ρ in the derivation step.

We say that (q, γ') is derivable from (p, γ) in n steps, written $(p, \gamma) \xrightarrow{n}_G (q, \gamma')$ iff $p = q$ and $\gamma = \gamma'$, when $n = 0$, or there are a picture r and an equivalence relation γ'' such that $(p, \gamma) \xrightarrow{n-1}_G (r, \gamma'')$ and $(r, \gamma'') \Rightarrow_G (q, \gamma')$. We use the abbreviation $(p, \gamma) \Rightarrow_G^* (q, \gamma')$ for a derivation with $n \geq 0$ steps.

Definition 8. The picture language defined by a grammar G (written $L(G)$) is the set of $p \in \Sigma^{**H}$ such that, if $|p| = (\ell, m, n)$, then

$$(S^{(\ell, m, n)}, coord(p) \times coord(p) \times coord(p)) \Rightarrow_G^* (p, \gamma) \quad (1)$$

where the relation γ is arbitrary. In short we write $S \Rightarrow_G^* p$.

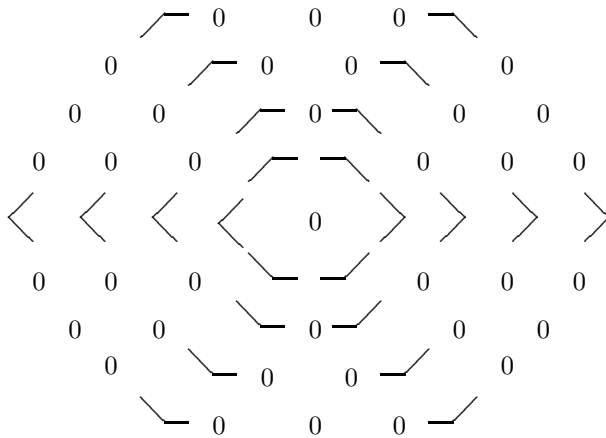
We note that the derivation starts with a S -picture isometric with the terminal picture to be generated, and with the universal equivalence relation over the coordinates. The equivalence relations computed by each step of (1) are called geminal relation. When writing examples by hand, it is convenient to visualize the equivalence classes of a geminal relation, by appending the same numerical subscript to the pixels of the application area rewritten by a derivation step.

Example 2. Let $G = (\Sigma \cup \{b\}, N, S, R)$, where

$\Sigma = \{ \nearrow, \searrow, \langle, \rangle, \swarrow, \nwarrow, 0 \}$, $N = \{S\}$, and R consists of the rules:

$$\begin{aligned}
 S &\rightarrow \begin{array}{|c|} \hline \diagup \quad \diagdown \\ \hline \end{array}; S \rightarrow \left\{ \begin{array}{|c|} \hline \begin{array}{cccccccc} 0 & S & S & S & S & S & S & 0 \end{array} \\ \hline \begin{array}{cccccccc} S & S & S & S & S & S & S & 0 \end{array} \end{array} \right\}, \\
 &\begin{array}{|c|} \hline \begin{array}{cccccccc} S & S & S & S & 0 & S & 0 & 0 & S & S \end{array} \\ \hline \begin{array}{cccccccc} S & S & 0, & 0 & S & S, & \langle & S & S, & S & S & S & S, \\ 0 & \swarrow & \searrow & 0 & 0 & S & S & S & 0 & 0 \end{array} \end{array} \\
 &\left. \begin{array}{|c|} \hline \begin{array}{cccccccc} S & S & S & 0 & 0 & S & S & S & S & S \end{array} \\ \hline \begin{array}{cccccccc} S & S & S, & S & S & 0, & 0 & S & S, & S & S & 0, & 0 & S & S \\ S & S & S & S & S & S & S & 0 & 0 & S \end{array} \end{array} \right\}
 \end{aligned}$$

A picture in $L(G)$ is:



and is obtained by repeated application of the variable size rule and fixed size rule.

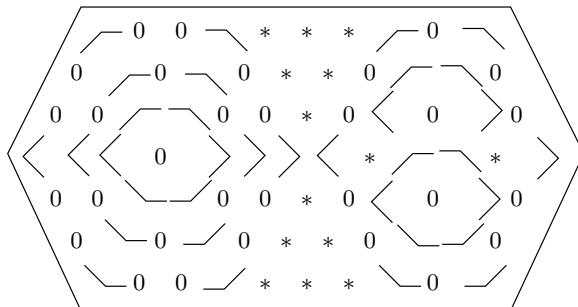
Example 3 (Dyck analogue). The next language can be defined by a sort of blanking. But since the terminals cannot be deleted without shearing the picture, we replace them with a character 'b' (blank or background).

To obtain the grammar, we add the following rules to the rules in Example 2.

$$S \rightarrow \left(\begin{array}{cccccc} S & S & * & * & X \\ S & S & S & * & X & X \\ S & S & S & S & X & X & X \\ S & S & S & * & X & X \\ S & S & * & * & X \end{array} \right) \left\| \left(\begin{array}{cccc} S & S & & \\ S & S & S & \\ * & S & S & * \\ X & X & X & \\ X & X & & \end{array} \right) \right.$$

$$X \rightarrow \left(\begin{array}{cc} S & S \\ S & S & S \\ S & S & \end{array} \right)$$

A picture in $L(G)$ is:



Now we give basic properties. For this, we require the notion of arrow head catenations. There are six arrow head catenations defined in [9]: upper left catenation, upper right catenation, left catenation, lower left catenation, lower right catenation, right arrow head catenations denoted by $\odot$, $\oslash$, $\ominus$, $\otimes$, $\otimes$, $\ominus$ respectively.

Theorem 1 (Closure Property). *The family $\mathcal{L}(HTRG)$ is closed under union, arrow head catenations, rotation about 180° , and projection.*

Proof. Consider two grammars $G_1 = (\Sigma \cup \{b\}, N_1, A, R_1)$ and $G_2 = (\Sigma \cup \{b\}, N_2, B, R_2)$. Suppose for simplicity that $N_1 \cap N_2 = \emptyset$ and $S \notin N_1 \cup N_2$. Then it is easy to show that the grammar $G = (\Sigma \cup \{b\}, N_1 \cup N_2 \cup \{S\}, S, R_1 \cup R_2 \cup R)$ where

$$\text{Union } U : R = \left\{ S \rightarrow \begin{pmatrix} A & A \\ A & A \\ A & A \end{pmatrix}, S \rightarrow \begin{pmatrix} B & B \\ B & B \\ B & B \end{pmatrix} \right\}$$

is such that $L(G) = L(G_1) \cup L(G_2)$.

$$\text{Concatenation : } \odot, \oslash, \ominus, \otimes, \otimes, \ominus$$

$$R = \left\{ S \rightarrow \begin{pmatrix} A & A & B & B \\ A & A & A & B & B \\ A & A & B & B \end{pmatrix} \right\}$$

is such that $L(G) = L(G_1) \odot L(G_2)$.

Similarly the other catenation cases are analogous.

Rotation about 180° : Construct the grammar $G = (\Sigma \cup \{b\}, N, A, R')$ where R' is such that, if $B \rightarrow t \in R_1$ is a type 1 rule, then $B \rightarrow t^R$ is in R' and if $B \rightarrow \omega \in R_1$ is a type 2 rule, then $B \rightarrow \omega'$ is in R' , with $t \in \omega \Rightarrow t^R \in \omega'$. It is easy to verify that $L(G) = L(G_1)_R$.

Projection π : Consider a grammar $G = (\Sigma \cup \{b\}, N, S, R)$ and a projection $\pi : \Sigma_1 \rightarrow \Sigma_2$. It is possible to build a grammar $G' = (\Sigma_2 \cup \{b\}, N', S, R')$ such that $L(G') = \pi(L(G))$. Indeed, let Σ'_1 be a set of new nonterminals corresponding to elements in Σ_1 . Then $N' = N \cup \Sigma'_1 \cup \{b\}$, $R' = \phi(R) \cup R''$, where

$$\phi : \Sigma_1 \cup \{b\} \times \{0\} \rightarrow \Sigma'_1 \cup \{b\} \times \{k\}$$

is the alphabetical mapping $\pi(a) = a'_k$, where k is a fixed unused updating index, and it is naturally extended to HTRG rules.

4 Comparison Results

We prove that the class of local hexagonal picture languages is strictly included in $\mathcal{L}(HTRG)$ and $\mathcal{L}(HTS) \subseteq \mathcal{L}(HTRG)$.

Theorem 2. $\mathcal{L}(HLOC) \subseteq \mathcal{L}(HTRG)$.

Proof. Consider a local hexagonal language over $\Sigma \cup \{b\}$ defined by the set of allowed hexagonal tiles Δ . Here ‘ b ’ stands for blank set.

Let $\Delta_0 = \left\{ \begin{array}{ccc|ccc} p_0 & q_0 & & p & q & \\ x_0 & y_0 & z_0 & x & y & z \in \Delta \\ u_0 & v_0 & & u & v & \end{array} \right\}$. Then an equivalent HTRG is $G = (\Sigma \cup \{b\}, \{S\}, S, R)$ where R is the set $\{S \rightarrow \theta/\theta \subseteq \Delta_0\}$.

Lemma 1. $\mathcal{L}(HLOC_{u,eq}) \subseteq \mathcal{L}(HTRG)$

Proof. Consider a local hexagonal picture language over Σ^{**H} defined, without boundaries by the sets of allowed tiles $\{\omega_1, \omega_2, \dots, \omega_n\}$, $\omega_i \subseteq \Sigma^{(2,2,2)^H}$. An equivalent grammar is $S \rightarrow \omega_1/\omega_2/\dots/\omega_n$.

We now consider the notions: HTS_{eq} and $HTS_{u,eq}$.

Definition 9. The hexagonal tiling systems HTS_{eq} and $HTS_{u,eq}$ are the same as a HTS, with the following respective changes;

- Replace the local language involved with $HLOC_{eq}(\{\omega_1, \omega_2, \dots, \omega_n\})$, where ω_i is a finite set of hexagonal tiles over Γ .
- Replace the local language involved with $HLOC_{u,eq}(\{\omega_1, \omega_2, \dots, \omega_n\})$, where ω_i is a finite set of hexagonal tiles over Γ . In $HTS_{u,eq}$ there is no boundary symbol $\#$.

Lemma 2. $\mathcal{L}(HTS_{eq}) \equiv \mathcal{L}(HTS)$

Proof. We first prove $\mathcal{L}(HTS) \subseteq \mathcal{L}(HTS_{eq})$. This is easy, because if we consider the hexagonal tile set ω of a HTS, by taking $\{\omega_1, \omega_2, \dots, \omega_n\} = P(\omega)$, (the power set) we obtain an equivalent HTS_{eq} . Next we have to prove that $\mathcal{L}(HTS_{eq}) \subseteq \mathcal{L}(HTS)$. In [7], the family of languages $\mathcal{L}(HLOC_{eq}(\Omega))$ where Ω is the set of tiles, is proved to be a proper subset of $\mathcal{L}(HTS)$. But $\mathcal{L}(HTS)$ is closed with respect to projection, and $\mathcal{L}(HTS_{eq})$ is the closure with respect to projection of $\mathcal{L}(HLOC_{eq}(\Omega))$. Therefore, $\mathcal{L}(HTS_{eq}) \subseteq \mathcal{L}(HTS)$.

Lemma 3. $\mathcal{L}(HTS_{u,eq}) \equiv \mathcal{L}(HTS_{eq})$.

Proof. First we prove $\mathcal{L}(HTS_{eq}) \subseteq \mathcal{L}(HTS_{u,eq})$.

Let $T = (\Sigma, \Gamma, (\omega_1, \omega_2, \dots, \omega_n), \pi)$ be a HTS_{eq} . For every hexagonal tile set ω_i separate its tiles containing the boundary symbol $\#$ (call this subset ω'_i) from the other tiles (ω''_i). This is $\omega_i = \omega'_i \cup \omega''_i$. Introduce a new alphabet Γ' and a bijective mapping $br : \Gamma \rightarrow \Gamma'$. We use symbols in Γ' to encode boundary, and new tile set δ_i to contain them: for every tile t in ω''_i , if there is a tile in ω'_i , which overlaps with t , then encode this boundary in a new tile t' and put it in the set δ_i .

Consider a $HTS_{u,eq}$, $T' = (\Sigma, \Gamma \cup \Gamma', \Omega, \pi')$ where π' extends π to Γ' as follows $\pi'(br(a)) = \pi'(a) = \pi(a)$, $a \in \Gamma$, and $ubr : \Gamma \cup \Gamma' \rightarrow \Gamma$ is defined as $ubr(a) = br^{-1}(a)$, if $a \in \Gamma'$, otherwise $= a$, and it is naturally extended to tiles and tile sets. Ω is the set

$$\{\omega/\omega \subseteq \omega''_i \cup \delta_i \wedge ubr(\omega) = \omega''_i \wedge \omega \cup \delta_i \neq \phi \wedge 1 \leq i \leq n\}$$

The proof of $L(T) = L(T')$ is obvious and is omitted.

To prove $\mathcal{L}(HTS_{u,eq}) \subseteq \mathcal{L}(HTS_{eq})$, let $T = (\Sigma, \Gamma, \{\omega_1, \omega_2, \dots, \omega_n\}, \pi)$ be a $HTS_{u,eq}$. To construct an equivalent HTS_{eq} , we introduce the boundary tile sets δ_i , defined as follows. For every tile,

$$\begin{array}{ccccccc}
 & d & e & & & & \\
 a & b & c & \in \omega_i, & \text{the following tiles are in } \delta_i : & & \\
 & f & g & & & & \\
 & \# & \# & \# & \# & d & \\
 \# & d & e, d & e & \#, \# & a & b, \\
 & a & b & b & c & \# & f \\
 & & & & e & \# & a & b & b & c \\
 & & & & b & c & \#, \# & f & g, f & g & \#. \\
 & & & & g & \# & \# & \# & \# & \#
 \end{array}$$

Consider a HTS_{eq} , $T' = (\Sigma, \Gamma, \Omega, \pi)$ where Ω is the set $\{\omega \cup \omega_i / \omega \subseteq \delta_i \wedge \omega \neq \phi \wedge 1 \leq i \leq n\}$. It is easy to show that $L(T) = L(T')$.

Theorem 3. $\mathcal{L}(HTS) \subseteq \mathcal{L}(HTRG)$.

Proof. It follows from theorem 1, lemma 1, lemma 2, lemma 3 and the fact that $\mathcal{L}(HTS_{u,eq})$ is the closure of $\mathcal{L}(HLOC_{u,eq})$ with respect to projection.

References

1. Brimkov, V.E., Barneva, R.: Analytical honeycomb geometry for raster and volume graphics. *The Computer Journal* 48(2), 180–199 (2005)
2. Cherubini, A., Crespi Reghizyi, S., Pradella, M., San Pietro, P.: Picture languages: Tiling systems versus tile rewriting grammars. *Theoretical Comp. Sci.* 356, 90–103 (2006)
3. Crespi Reghizyi, S., Pradella, M.: Tile rewriting grammars. In: Ésik, Z., Fülöp, Z. (eds.) *DLT 2003*. LNCS, vol. 2710, pp. 206–217. Springer, Heidelberg (2003)
4. Crespi Reghizyi, S., Pradella, M.: Tile rewriting grammars and picture languages. *Theoretical Comp. Science* 340, 257–272 (2005)
5. Dersanambika, K.S., Krithivasan, K., Martin-Vide, C., Subramanian, K.G.: Hexagonal Pattern Languages. In: Klette, R., Žunić, J. (eds.) *IWCIA 2004*. LNCS, vol. 3322, pp. 52–64. Springer, Heidelberg (2004)
6. Deutsch, E.S.: Thinning algorithms on rectangular, hexagonal and triangular arrays. *Communications of the ACM* 15(9), 827–837 (1972)
7. Giammarresi, D., Restivo, A.: Recognizable Picture Languages. *Internat J. Pattern Recogn. Artif. Intell.* 6(2 & 3), 241–256 (1992)
8. Luszak, E., Rosenfeld, A.: Distance on a hexagonal grid. *IEEE Transactions on Computers* 25(5), 532–533 (1968)
9. Siromoney, G., Siromoney, R.: Hexagonal arrays and rectangular blocks. *Computer Graphics and Image Processing* 5, 353–381 (1976)
10. Subramanian, K.G.: Hexagonal Array Grammars. *Computer Graphics and Image Processing* 10, 388–394 (1979)
11. Sweety, F., Thomas, D.G., Kalyani, T.: Collage of Hexagonal Arrays. In: Bebis, G., et al. (eds.) *ISVC 2008, Part I*. LNCS, vol. 5358. Springer, Heidelberg (2008)
12. Wuthrich, C.A., Stucki, P.: An algorithm comparison between square and hexagonal-based grids. *Graphical Models and Image Processing* 53(4), 324–339 (1991)